

Math-m-addicts group Y+: Distributive property.

“Distributive Property”: $a \cdot (b + c) = ab + ac$.

The distributive property can be used to expand parentheses as well as factor.

Usually, expansion is easy and automatic while factoring requires practice and sometimes highly non-trivial insights.

Example 1: Using distributive property, expand: $(a + b) \cdot (a + c)$.

$$(a + b) \cdot (a + c) = (a + b) \cdot a + (a + b) \cdot c = a^2 + ab + ac + bc.$$

Pretty easy, right?

When opening parentheses in a product of two sums, you multiply EVERY term in one sum by EVERY term in the other sum. If the first sum contains m terms and the second sum contains n terms then, before reducing like terms, the product will contain $m \cdot n$ terms. In example 1, you should expect to get $2 \cdot 2 = 4$ terms.

Problem 1. Fully expand the following expressions. Collect like terms and simplify.

Example: $(x + 1)(x + 2) = x^2 + 3x + 2$

- a) $(x - 1) \cdot x \cdot (x + 1)$
- b) $(x + 6)(x - 1) - (x - 6)(x + 1)$
- c) $a(b - c) + b(c - a) + c(a - b)$
- d) $(a + 1)(b - 1)$
- e) $(a + 3)(a - 3)$
- f) $(b + 2)^2 + (b - 2)^2$
- g) $(a + b)^2$
- h) $(a - b)^2$
- i) $(a + b) \cdot (a - b)$
- j) $(a + b)^3$
- k) $(a + b)^3 - (a - b)^3$
- l) $(a + b) \cdot (a^2 - ab + b^2)$.
- m) $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$. Do it.

Example 2: Let a, b, c be positive integers such that $a^2 + ab + ac + bc = 94$. Find the largest possible value of c .

Note that, as a standalone problem, this is not nearly as easy as Example 1. However, you should immediately recognize the given expression as the expansion from Example 1. Thus:

$$a^2 + ab + ac + bc = (a + b) \cdot (a + c) = 94 = 2 \cdot 47.$$

Because a, b, c are positive integers, both $(a + c)$ and $(b + c)$ are at least 2. There is only one way to factor 94 as a product of two integers greater than 1, that is $94 = 2 \cdot 47$.

Therefore, either $a = b = 1, c = 46$ or $a = c = 1, b = 46$. Hence the largest possible value of c is 46.

Problem 2. a, b, c, d are positive integers satisfying $ab + bc + cd + ad = 35$. Find $a + b + c + d$.

Identities: Expansions g-i of Exercise 1 are EXTREMELY common. You must know these:

$$\text{I1: } (a + b)^2 = a^2 + b^2 + 2ab.$$

$$\text{I2: } (a - b)^2 = a^2 + b^2 - 2ab$$

$$\text{I3: } (a - b)(a + b) = a^2 - b^2$$

Problem 3. Compute without a calculator, using I1-I3:

a) $43 \cdot 36 + 43 \cdot 64.$

b) $53 \cdot 47$

c) $43^2 + 37^2 + 86 \cdot 37$

d) $63^2 + 37^2$

e) $113^2 - 87^2.$

How to work with scary fractions:

Frequently one needs to put fractions with variables in them to a common denominator. This may seem intimidating but it is exactly the same as what you would do with numbers. For example:

$$\frac{1}{n} + \frac{1}{n(n-2)} = \frac{n-2}{n(n-2)} + \frac{1}{n(n-2)} = \frac{n-1}{n(n-2)}$$

Problem 4: Simplify the following fractions by factoring and canceling terms. You may assume that the factors you are canceling are not zero, for simplicity. Do not use a calculator for any of the problems and AVOID computing anything by hand as much as possible.

Example: $\frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1} = x+1. \quad x \neq 1.$

a) $\frac{52^2+48^2+2 \cdot 52 \cdot 48}{52^2-48^2}$

b) $\frac{x^2+3x+2}{x^2+2x+1}$

c) $\frac{x^3-x}{x^2-x}$

d) $\frac{2^{16}-1}{2^8-1}$

e) $\frac{2^{16}-1}{17 \cdot (2^8+1)}$

f) $\frac{9^5-4}{241}$

g) $\frac{1+\frac{1}{x-1}}{1+\frac{1}{x^2-1}}$

Problem 5. (more fractions practice): Prove that for any $n \neq 0, 1$:

$$\frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$$

Problem 6. Basic factorial manipulation:

a) Prove that $n \cdot n! = (n + 1)! - n!$

b) Prove Pascal's identity: $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$ for positive integers n, k with $k \leq n$.